



Link between household income and elite college admission

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Introduction to our research question

- We want to know whether students from higher-income families are more likely to attend elite colleges than students from lower-income families.
- Our main focus is on the relationship between parent income percentile bins and attendance at “elite” college tiers.
- This question matters because it connects to equity and access in higher education: who actually ends up at the most selective schools?
- We’re using observational data, so we’re looking for patterns and associations, not proving strict causation.

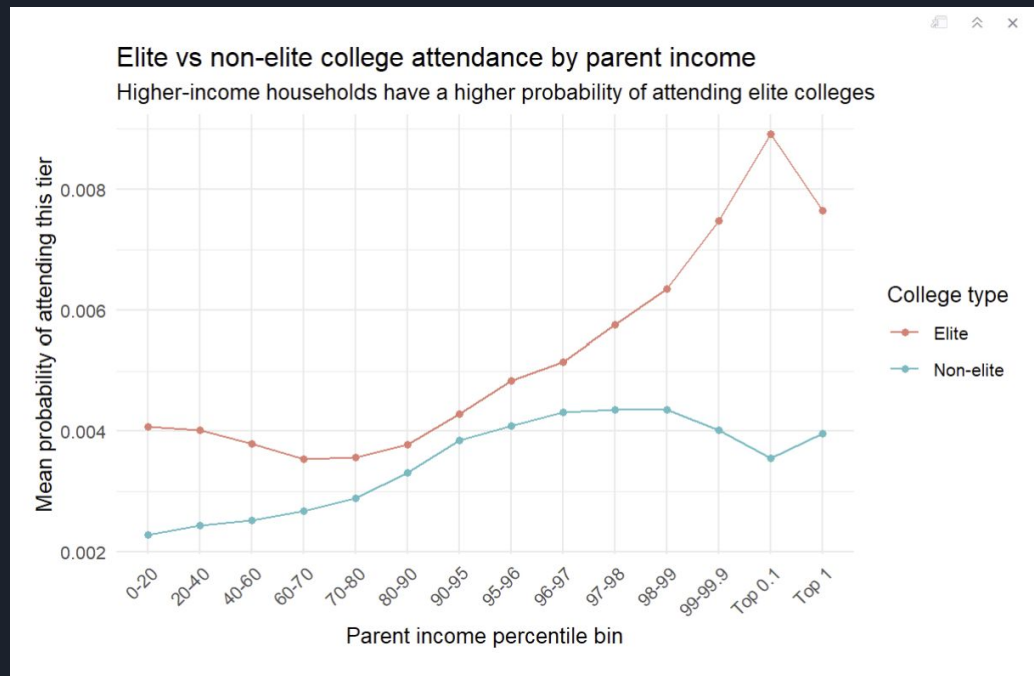


Overview of our data

- Dataset:
<https://cmustatistics.github.io/data-repository/social/elite-college-admissions.html>
with information on 139 colleges and 1,900+ college-income-bin combinations.
- Key variables we use:
 - `par_income_lab`: parent income percentile bins (e.g., 0–20, 20–40, ..., 99–99.9, Top 0.1).
 - `attend`: probability that a student from that income bin attends that college.
 - `tier_name`: describes the selectivity tier of the college (Ivy Plus, highly selective, etc.).
- We define “elite” colleges as: Ivy Plus, Highly selective private, Highly selective public, and Other elite schools (public and private).

Analysis of data

- For low- and middle-income bins (0–20 up to ~80–90), the elite line is a bit above the non-elite line, but both are pretty flat and low. That means students from these income groups have a small chance of attending *any* given college tier, and the gap between elite vs non-elite isn't huge yet.
- Starting around the 90–95 bin, both lines start to climb, but the elite line climbs much faster. By the top income groups (98–99, 99–99.9, Top 0.1, Top 1), the elite attendance probability jumps sharply, while the non-elite line rises more slowly and even dips slightly before a small bump at Top 1. This widening gap means high-income students are increasingly concentrated in elite colleges compared to non-elite ones.
- Overall, the shape of the graph shows a clear positive relationship between parent income and elite college attendance: as income goes up, the probability of attending an elite college increases more steeply than the probability of attending a non-elite college.





Hypothesis Test

- To move beyond just a plot, we set up a formal hypothesis test using a Pearson correlation(P-test)
- Null hypothesis (H_0): $\rho = 0 \rightarrow$ there is no linear correlation between parent income bin and attendance probability at elite colleges.
- Alternative hypothesis (H_1): $\rho > 0 \rightarrow$ higher income bins are associated with higher attendance probabilities at elite colleges.
- We ran `cor.test(income_order, attend, alternative = "greater")` to get the estimated correlation and p-value.

Interpreting our results

- We tested whether parent income bin is positively related to attendance at elite colleges
- The sample correlation is about 0.27, which means there is a moderate positive linear relationship: higher income bins tend to have higher attendance probabilities.
- The p-value is essentially 0 ($3.7\text{e-}26$), far below 0.05, so we reject the null hypothesis of no correlation and accept that the true correlation is greater than 0.
- The 95% confidence interval for the true correlation runs from about 0.23 up to 1 (because it's one-sided), so we're confident the real correlation is at least around 0.23 and clearly positive.
- In summary, the hypothesis test shows that students from higher-income households are significantly more likely to attend elite colleges than those from lower-income households.

Pearson's product-moment correlation

```
data: income_order and attend
t = 10.714, df = 1496, p-value < 2.2e-16
alternative hypothesis: true correlation is greater than 0
95 percent confidence interval:
 0.2270053 1.0000000
sample estimates:
      cor
0.2669443
```

```
Estimated correlation (rho): 0.267
p-value: 3.7e-26
```



Conclusion

- Our analysis provides clear evidence that higher parent income is linked to a greater chance of attending an elite college.
- Both the visualization and the hypothesis test point in the same direction: attendance probabilities rise as we move from low-income to top-income bins, especially for elite tiers.
- This doesn't prove that income alone "causes" elite admissions, but it shows that income and elite attendance move together in a statistically significant way
- Limitations: no individual test scores, no controls for other factors (race, high school quality, geography), and the data are aggregated.
- Possible next steps: add more variables, run regression models, or compare results across different subgroups to see where the income–elite gap is largest